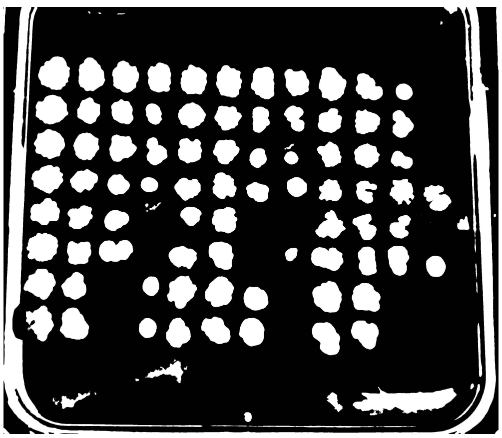
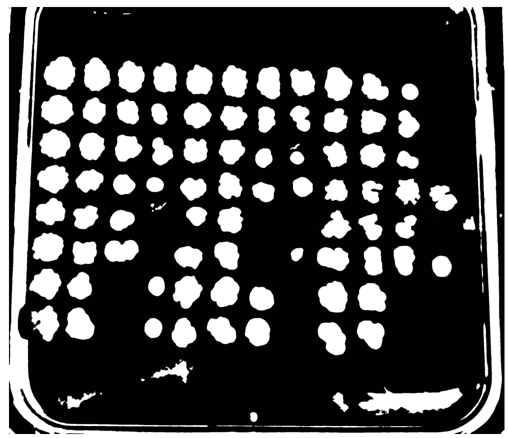
Ecoli-1_NoDrug_IMG_3310.JPG Log

Preview Image

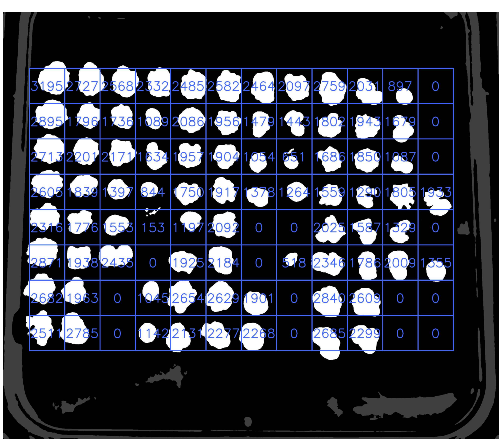
Filename: Ecoli-1_NoDrug_IMG_3310.JPG
Additive: None
X Dilution: 10
Y Dilution: 2
Strains: Msm, Msm, Msm
Column Indexes: [[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11]]
Gap Between Strains: False
Threshold: 2
Minimum Area: 15
Block Size: 179



Tool Usage: Red(+) Blue(-))



Binary Image



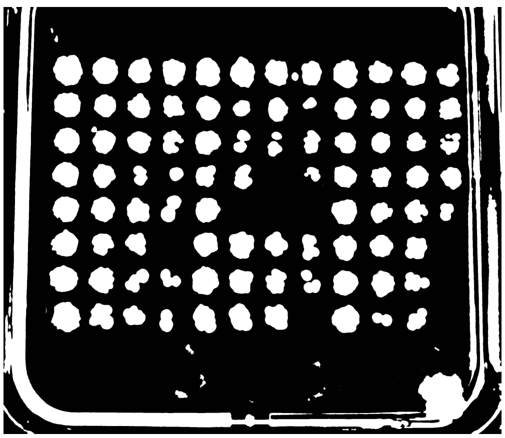
Grid Image

Contours Image

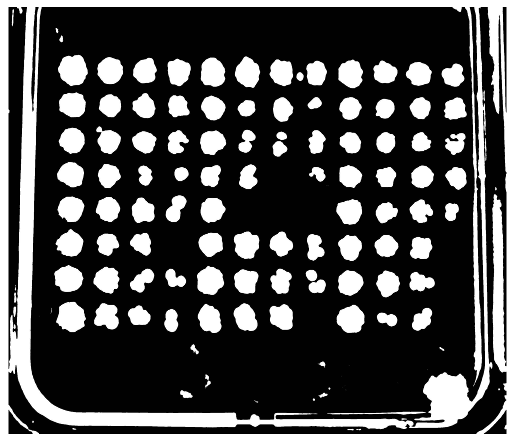
Ecoli-3_NoDrug_IMG_3312.JPG Log

Preview Image

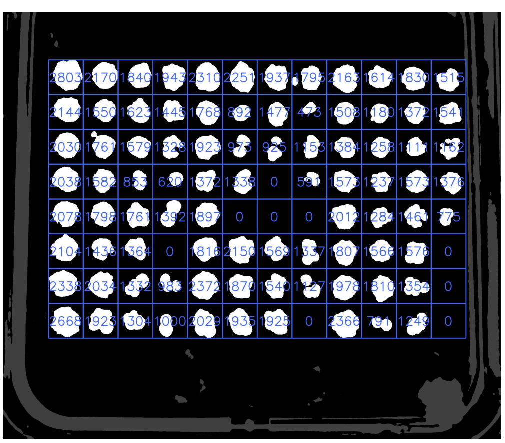
Filename: Ecoli-3_NoDrug_IMG_3312.JPG
Additive: None
X Dilution: 10
Y Dilution: 2
Strains: Msm, Msm, Msm
Column Indexes: [[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11]]
Gap Between Strains: False
Threshold: 6
Minimum Area: 15
Block Size: 179



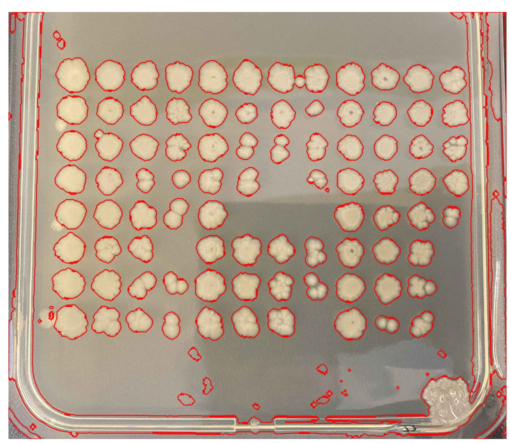
Tool Usage: Red(+) Blue(-))



Binary Image



Grid Image

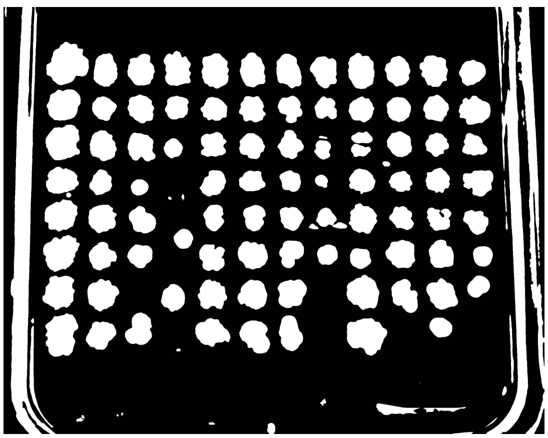


Contours Image

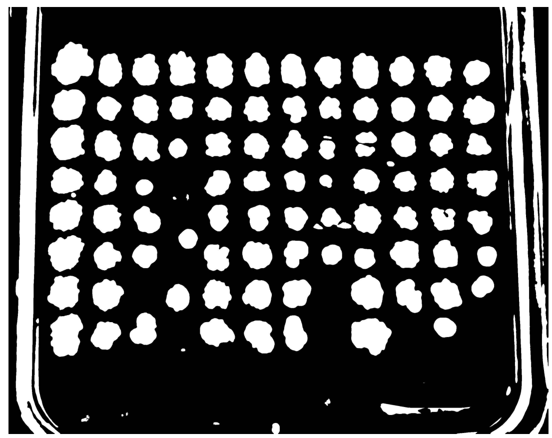
Ecoli-4_Kanamycin_IMG_3315.JPG Log

Preview Image

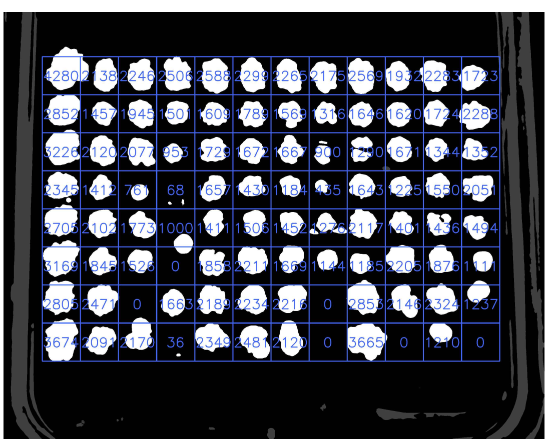
Filename: Ecoli-4_Kanamycin_IMG_3315.JPG
Additive: Kanamycin
X Dilution: 10
Y Dilution: 2
Strains: Msm, Msm, Msm
Column Indexes: [[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11]]
Gap Between Strains: False
Threshold: 4
Minimum Area: 15
Block Size: 179



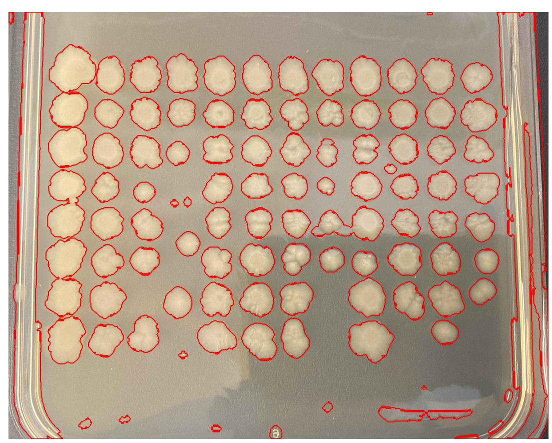
Tool Usage: Red(+) Blue(-))



Binary Image

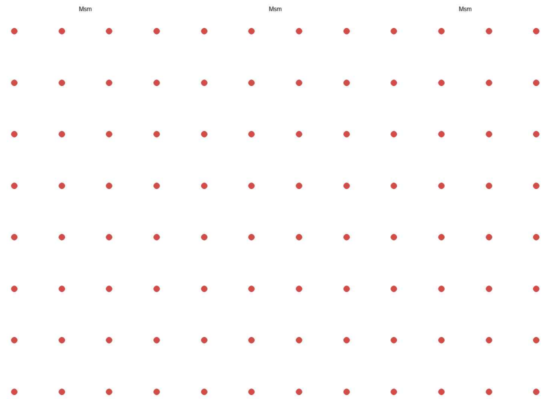


Grid Image



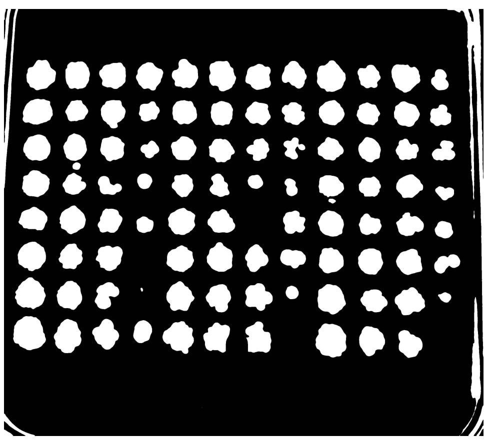
Contours Image

Ecoli-2_Kanamycin_IMG_3311.JPG Log

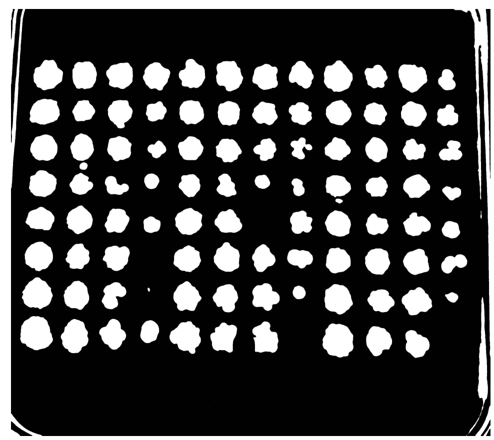


Preview Image

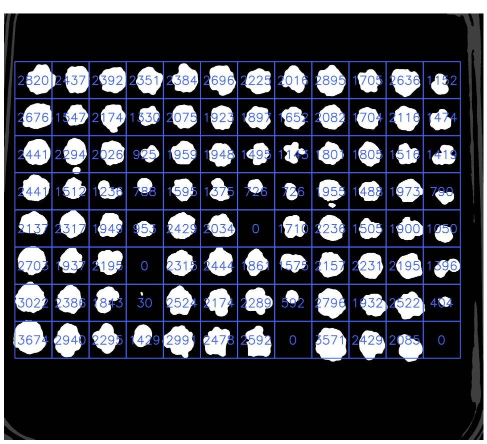
Filename: Ecoli-2_Kanamycin_IMG_3311.JPG
Additive: Kanamycin
X Dilution: 10
Y Dilution: 2
Strains: Msm, Msm, Msm
Column Indexes: [[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11]]
Gap Between Strains: False
Threshold: 7
Minimum Area: 15
Block Size: 179



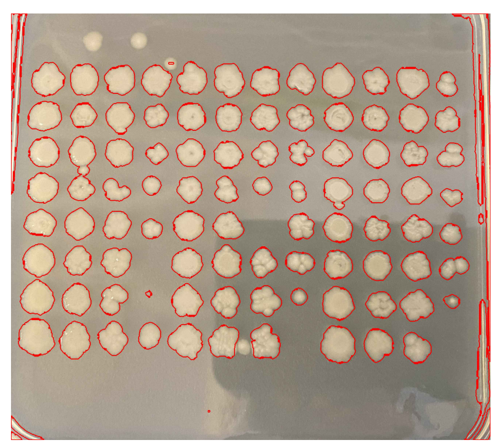
Tool Usage: Red(+) Blue(-))



Binary Image

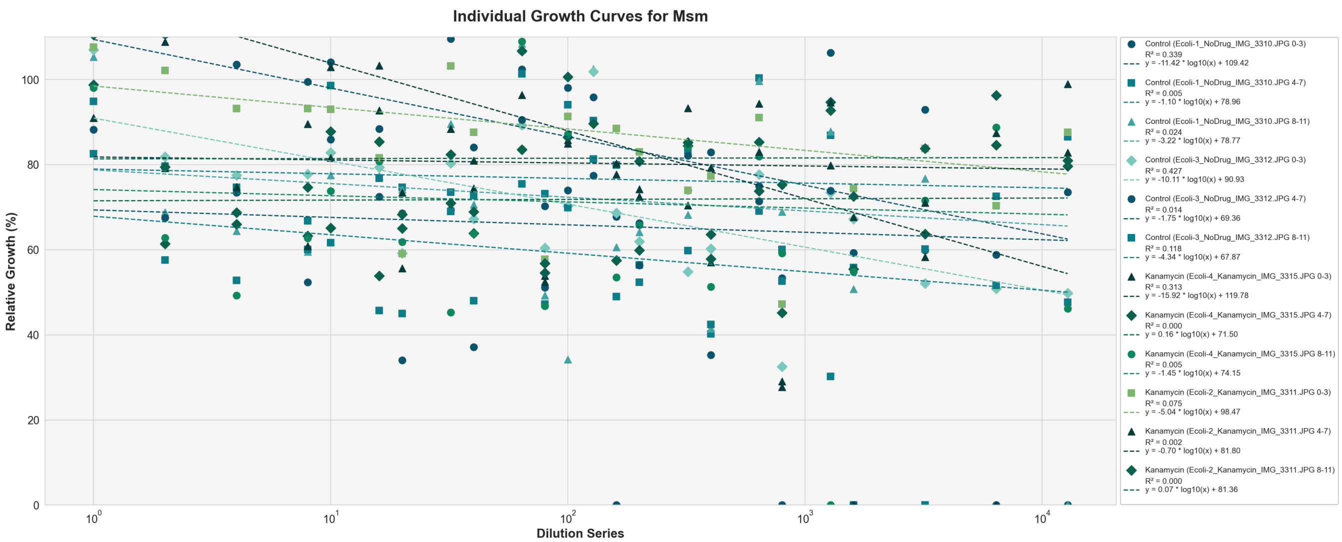


Grid Image



Contours Image

SPOTPLOTTER
AUTOMATED SPOTTING ASSAY QUANTIFICATION



	Control (Ecoli-1_NoDrug_IMG_3310.JPG 0-3)	Control (Ecoli-1_NoDrug_IMG_3310.JPG 4-7)	Control (Ecoli-1_NoDrug_IMG_3310.JPG 8-11)	Control (Ecoli-3_NoDrug_IMG_3312.JPG 0-3)
Formula	y = -11.42 * log10(x) + 109.42	y = -1.10 * log10(x) + 78.96	y = -3.22 * log10(x) + 78.77	y = -10.11 * log10(x) + 90.93
Slope	-11.42	-1.10	-3.22	-10.11
Intercept	109.42	78.96	78.77	90.93
R-squared	0.339	0.005	0.024	0.427
y-cut	109.42	78.96	78.77	90.93
x-cut	3.80e+09	5.65e+71	3.15e+24	9.93e+08
x_at_y50	1.59e+05	2.07e+26	8.86e+08	11216.88

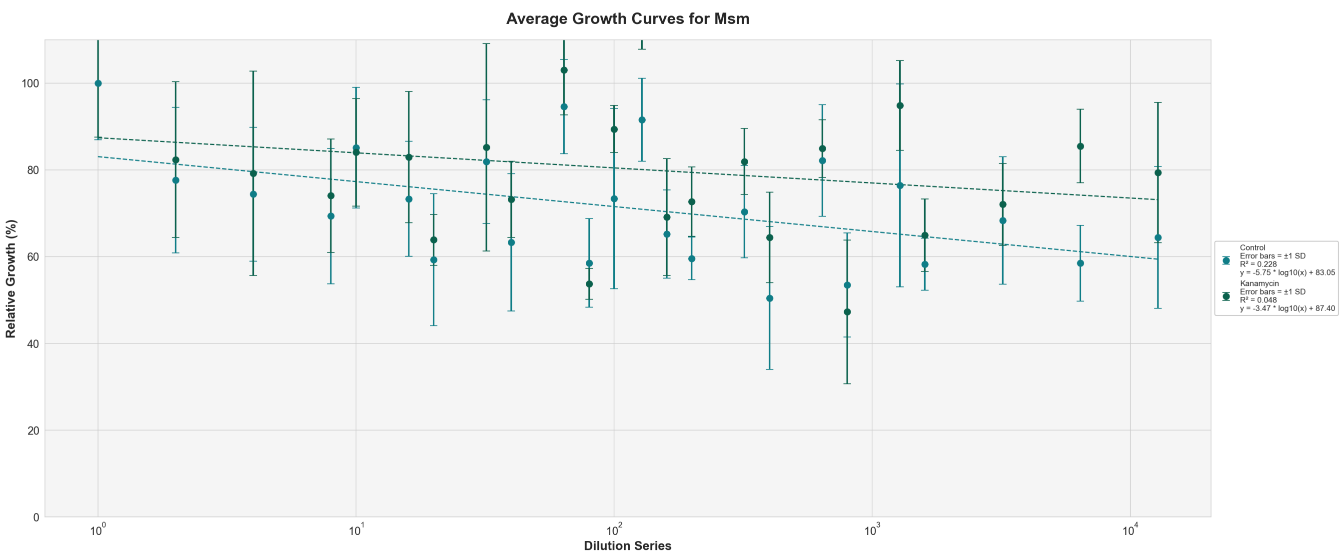
	Control (Ecoli-3_NoDrug_IMG_3312.JPG 4-7)	Control (Ecoli-3_NoDrug_IMG_3312.JPG 8-11)	Kanamycin (Ecoli-4_Kanamycin_IMG_3315.JPG 0-3)	Kanamycin (Ecoli-4_Kanamycin_IMG_3315.JPG 4-7)
Formula	y = -1.75 * log10(x) + 69.36	y = -4.34 * log10(x) + 67.87	y = -15.92 * log10(x) + 119.78	y = 0.16 * log10(x) + 71.50
Slope	-1.75	-4.34	-15.92	0.16
Intercept	69.36	67.87	119.78	71.50
R-squared	0.014	0.118	0.313	0.000
y-cut	69.36	67.87	119.78	71.50
x-cut	4.25e+39	4.20e+15	3.33e+07	0.00
x_at_y50	1.15e+11	13006.07	24131.92	0.00

	Kanamycin (Ecoli-4_Kanamycin_IMG_3315.JPG 8-11)	Kanamycin (Ecoli-2_Kanamycin_IMG_3311.JPG 0-3)	Kanamycin (Ecoli-2_Kanamycin_IMG_3311.JPG 4-7)	Kanamycin (Ecoli-2_Kanamycin_IMG_3311.JPG 8-11)
Formula	y = -1.45 * log10(x) + 74.15	y = -5.04 * log10(x) + 98.47	y = -0.70 * log10(x) + 81.80	y = 0.07 * log10(x) + 81.36
Slope	-1.45	-5.04	-0.70	0.07
Intercept	74.15	98.47	81.80	81.36

R-squared	0.005	0.075	0.002	0.000
y-cut	74.15	98.47	81.80	81.36
x-cut	1.58e+51	3.52e+19	1.13e+116	0.00
x_at_y50	4.71e+16	4.18e+09	1.32e+45	0.00

SPOTPLOTTER

AUTOMATED SPOTTING ASSAY QUANTIFICATION



	Control	Kanamycin
Formula	y = -5.75 * log10(x) + 83.05	y = -3.47 * log10(x) + 87.40
Slope	-5.75	-3.47
Intercept	83.05	87.40
R-squared	0.228	0.048
y-cut	83.05	87.40
x-cut	2.72e+14	1.45e+25
x_at_y50	5.55e+05	5.85e+10

SPOTPLOTTER

AUTOMATED SPOTTING ASSAY QUANTIFICATION

This report was generated by **Spotplotter** version **1.0** on **2024-11-25**

Spotplotter was created by **Holly Lewis** with supervision from **R Verrinder** and **Dr. M Mason** as BSc (Eng) final year project submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical and

Computer Engineering in the Department of Electrical Engineering at the University of Cape Town.

To read the full report please see: *GITHUB LINK*

Formula: The formula represents the linear regression equation that models determined using the by linregress function from scipy.stats which determines the relationship between the logarithm of the dilution series and relative growth. It follows the form: $y = m \cdot \log10(x) + b$

Normalization: The relative growth values were normalized to a baseline to make the results comparable across different conditions. The quantified values of each spot were divided by the average value of the first spot in the -ATP for each strain.

The slope (m): indicates the rate of change in relative growth as the dilution series increases (on a logarithmic scale). A steep slope indicated that the growth has a faster knockdown as the dilution changes. A shallow slope indicates that the growth is more stable across dilutions.

The intercept (b) and y-cut is the relative growth when the solution is not diluted.

The R-squared value measures how well the linear regression line fits the data, where 1 represents a perfect fit and 0 represents no relationship. Higher R^2 values indicates that the growth follows a linear relationship.

The X-cut refers to the point where the regression line crosses the x-axis, indicating the dilution value at which the relative growth would theoretically be zero (no growth).

X at Y = 50: This value represents the dilution series value when the relative growth is 50% i.e., the knockdown is 50% in comparison to the -ATP series.